Smart Energy Farm: Solar & Wind Energy

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INTRODUCTION- Idea

1. **Real-Time IoT Control:** Implemented bidirectional communication using HiveMQ (MQTT) to visualize live telemetry (Voltage/Current) and remotely control farm assets (Wind & Solar Farm).

2. **AI-Driven Power Prediction:** Integrated a Python-Flask backend with Random Forest ML models to accurately predict power generation based on real-time environmental inputs (Wind Speed & Irradiation).

3. **Advanced Data Analytics:** Features automated parsing and merging of large-scale CSV datasets (50k+ rows) to generate historical insights on power efficiency and seasonal trends.

INTRODUCTION- Hardware & Software Components

Hardware Components:

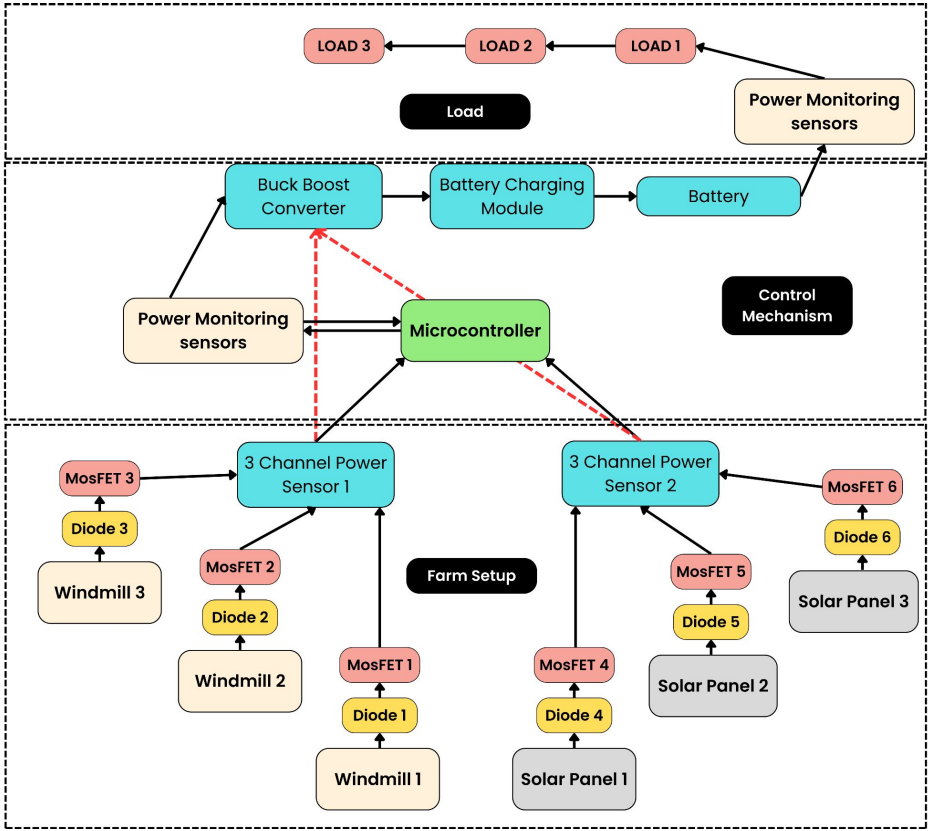
1. Setup - Solar Panel, Servo Motor, 3D Printed Wind Turbine and Dual-Axis Solar panel case, Battery, Motor.
2. uC - Arduino Uno R4, ESP32- S3.
3. Sensors - LDR, INA 226 (Current Sensor).

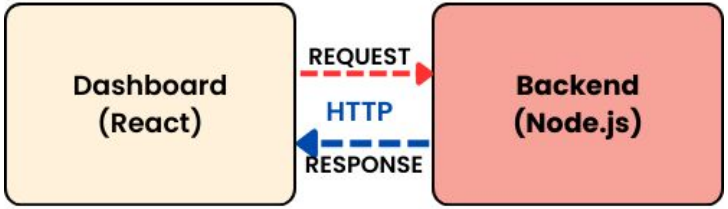
Software Components:

1. Cloud/Broker - HiveMQ.
2. Communication Protocols - MQTT.
3. Frontend: Done using React and TailwindCSS, for the data visualization and monitoring
4. Backend - A server (Python with Flask) to Run the ML model to be integrated with Frontend.

AI & ML: Predictive Analysis of the energy production for both Wind and Solar Energy.

BLOCK DIAGRAM- 1





ML MODEL

1. **Random Forest Regression Model** used in our project uses an ensemble of multiple decision trees, combining their outputs to produce accurate and stable predictions even in noisy real-world datasets.

2. It captures **nonlinear relationships between** the input and output power, making it ideal for SCADA-based wind turbine and solar inverter modeling.

3. The model is **highly robust**, reduces overfitting through averaging, and delivers **reliable performance** suitable for deployment in real-time digital twin systems.

ML MODEL

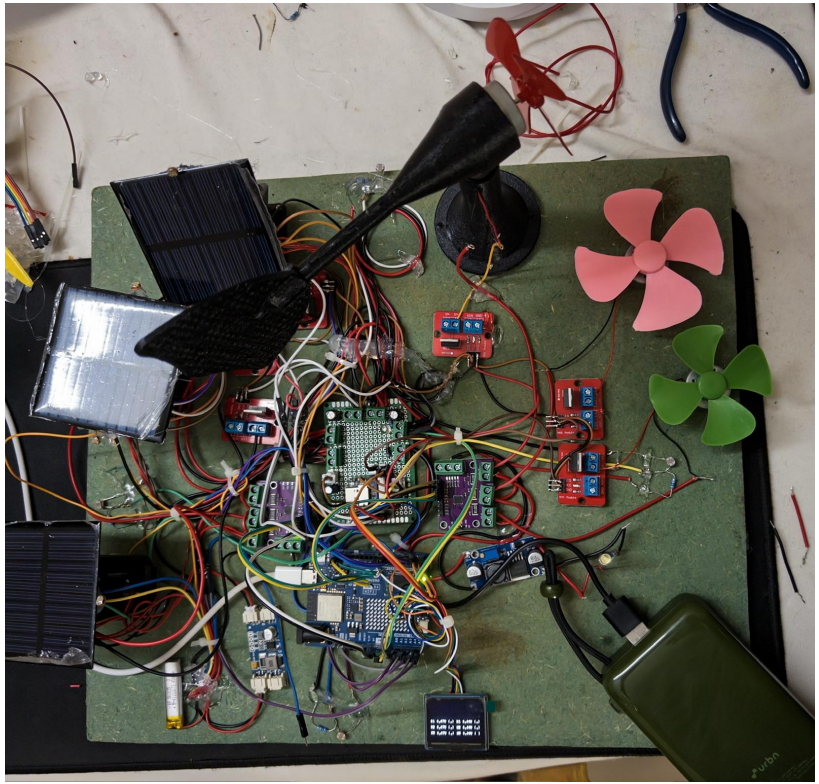
Wind Farm Model:

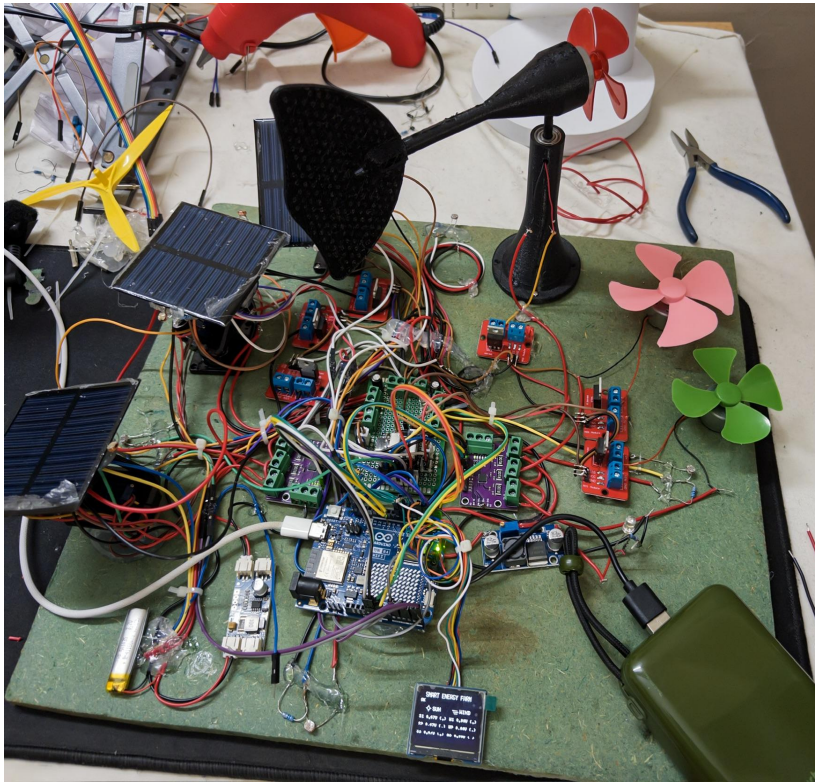
- Training Set Accuracy: 94.75% (R^2 = 0.9475)
- Testing Set Accuracy: 89.20% (R^2 = 0.8920)
- Model Generalization: Excellent

Solar Farm Model:

- Training Set Accuracy: 99.16% (R^2 = 0.9916)
- Testing Set Accuracy: 99.09% (R^2 = 0.9909)
- Model Generalization: Excellent

WORKING- Hardware





WORKING- Software

THANK YOU!